



Long-term outcomes of 47 patients with aggressive fibromatosis of the chest treated with surgery

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Abstract

Aims: The purpose of the study was to review a large series of chest aggressive fibromatosis (AF) cases with an emphasis on the outcomes of different degrees of resection and the value of postoperative radiotherapy.

Materials and methods: The records of patients with chest AF treated at our hospital from 1982 to 2014 were retrospectively reviewed. Recurrence rates and non-disease survival (NDS) times were compared between the R0, R1, and R2 resection groups.

Results: Forty-seven cases of chest AF were treated during the study period (21 men, 26 women), with an average age at diagnosis of 40 years (range, 9–77 years). One patient died before surgery, and 46 patients received a total of 85 resections. Forty-one patients had complete follow-up data, and the average follow-up time was 125.6 months (range, 11–524 months). Recurrence rates were 6.7%, 92.9%, and 100% for the R0, R1, and R2 resection groups, respectively, and the R0 recurrence rate was significantly lower than the R1 and R2 rates (both P values < 0.001). The NDS time of the R0, R1, and R2 groups was 80.3 ± 64.8 , 23.6 ± 38.7 , and 9.8 ± 10.8 months, respectively; the NDS time of the R0 group was significantly longer than that of the R1 and R2 groups (both P values < 0.01). Within each resection type, no significant differences were found in the recurrence rates of patients having surgery alone compared with those receiving surgery and radiotherapy (all P values > 0.05).

Conclusion: R0 resection is the most effective treatment for chest AF. Postoperative radiotherapy did not reduce the recurrence rate.

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Keywords: Aggressive fibromatosis; Desmoid; Reconstruction; Chest wall defect

Introduction

Aggressive fibromatosis (AF), also known as desmoid tumor or desmoid-type fibromatosis, was defined by the World Health Organization (WHO) in 2002 as a clonal fibroblast proliferative tumor occurring in deep tendon membrane tissues.¹ AF is characterized by abundant collagen fibers with involvement of the skeletal muscle aponeurosis and fascia; it has a tendency towards progressive infiltration of surrounding tissues and local recurrence without the ability to metastasize.¹ Its biological behavior is classified between that of fibroblastoma and fibrosarcoma.¹ AF accounts for 3% of soft tissue tumors, and

0.03% of all malignant tumors; in the United States, the incidence is 2–4 cases/100,000 persons yearly.² The disease occurs primarily in younger age groups, and there is no specific gender difference. Approximately 5% of tumors are multifocal, and the mortality is about 8%.²

Surgery is the primary therapy for resectable cases of chest AF, and after surgery alone, local recurrence rates range between 20% and 80%.^{2,3} Traditionally, the goal of surgery has been R0 resection; however, this is difficult in many cases of chest AF.^{4,5} Some authors have suggested less aggressive surgery and postoperative radiotherapy to improve the local control rate; however, the group of patients for whom this is suitable therapy remains a matter of debate.^{4,5} Other therapies, including irradiation, hormonal therapy, chemotherapy, and receptor tyrosine kinase

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inhibition, can be successful in stabilizing growth or shrinking lesions, and although no single therapy is effective in all cases, different therapies may be of value in certain pathological subgroups.⁶

AF is a rare disease for which there is a lack of prospective randomized studies; additionally, there are no standardized treatment guidelines,^{7,8} and the literature regarding chest AF is mostly case reports. Thus, the purpose of this study is to report a large series of chest AF cases treated at our institution with an emphasis on the outcomes of different degrees of resection and the value of postoperative radiotherapy.

Materials and methods

In this study, the records of patients diagnosed with chest AF and treated at Peking Union Medical College Hospital from December 1982 to May 2014 were retrospectively reviewed. This study was approved by the Institutional Review Board of the hospital, and because of the retrospective nature of the analysis, the requirement for the patients' informed consent was waived.

Data extracted from the medical records included patient age, gender, clinical history, surgical procedure, extent of resection, histopathological and immunohistochemical results, the use of radiotherapy, and outcome data. Patients were followed up every 6 months for the first 5 years after surgery and yearly thereafter. Postoperative follow-up consisted of outpatient examinations and annual ultrasound or magnetic resonance imaging (MRI) examinations.

All pathological specimens were fixed in 10% neutral-buffered formalin, embedded in paraffin wax, stained with hematoxylin and eosin (H&E), and examined under a fluorescence microscopy (Olympus BX51, Japan). For cases in which a diagnosis of AF was unclear, immunohistochemical staining was performed to distinguish a variety of tumors, such as smooth muscle tumors, neurogenic tumors, synovial sarcoma, fibrosarcoma, mesothelioma, solitary fibrous tumors, nodular fasciitis, malignant fibrous histiocytoma, and gastrointestinal stromal tumors.

Statistical analysis

Continuous data were reported as mean $\pm$ standard deviation (SD) or range, while categorical data were presented as number and percentage. Three surgical groups were examined and compared: R0, R1, and R2 resection groups. One-way ANOVA with Tukey's honest significant difference post-hoc test was used to compare continuous means when there were more than two groups, while a t-test was used for two-group comparisons. Chi-square tests and Fisher's exact tests, as appropriate, were used for comparisons of categorical data. Kaplan–Meier survival analysis and log-rank tests were used to compare the recurrence rate and non-disease survival time (NDS; time from surgery to first relapse) among the three resection

groups. Statistical analyses were done with SPSS, version 18, with an alpha level of 0.05.

Results

Patient clinical data and treatment

A total of 47 cases of chest AF were included in the analysis, and the clinical features are summarized in Table 1. These 47 cases accounted for 4.7% of chest soft tissue tumors and 18.7% of all AF cases diagnosed at our center during the study period. Twenty-six patients were diagnosed at our hospital, while 21 were relapses that were initially diagnosed and treated at other hospitals. There were 21 men (44.7%) and 26 women (55.3%), and the average age at diagnosis was 40 years (range, 9–77 years). There were 4 patients (8.5%) with multiple lesions, 2 of whom were females with familial adenomatous polyposis, and the other patients had single lesions.

A 77-year-old woman with giant chest-abdominal AF died before surgery due to respiratory failure. The other 46 patients received a total of 85 chest tumor resections: 28 R0, 24 R1, and 33 R2. There were 23 patients who had their initial surgery at our hospital; 14 of them had R0 resections. Twenty-five patients received single resections, and 21 required multiple resections. Artificial materials were used intraoperatively to repair the chest defects

Table 1
Clinical features of chest aggressive fibromatosis.

Presenting symptoms	Pain and numbness gradually developed in over half of the patients Upper limb swelling and numbness if the subclavian axillary region was involved, and craniofacial swelling if the upper mediastinum was involved
Examination	Typically, a non-tender mass with an unclear boundary. The larger the mass, the worse the mobility Approximately 38% of recurrent cases were a hard mass beneath the scar and were associated with pain and rapid growth
Imaging studies	Ultrasound, computed tomography (CT) and/or magnetic resonance imaging (MRI) revealed a soft tissue mass on the chest wall with uniform density and an unclear boundary. The average diameter was 6.9 cm (maximum diameter, 27 cm) In 17 patients who received a bone scan, radionuclide concentration in the lesion was observed in 8 patients, and bone invasion was pathologically confirmed in two cases
Involved areas	Chest wall muscles and ribs, 29 cases (61.7%); scapular area, 10 cases (21.3%); clavicular area and thoracic inlet, 10 cases (21.3%); thoracic outlet, 8 cases (17%); breast, 6 cases (12.8%); intrathoracic area, 5 cases (10.6%); paravertebral area, 5 cases (10.6%); mediastinum, 4 cases (8.5%); diaphragm, 3 cases (6.4%)

in 11 cases, and musculocutaneous flaps were used in 8 cases. There were 19 patients who received 55-Gy radiotherapy after surgery due to positive surgical margins or due to disease recurrence after complete resection. A total of five patients received chemotherapy as part of their treatment, and in 1 case, prophylactic chemotherapy was administered. Table 2 shows a summary of the surgical approaches to reconstruction, and the distribution of tumor location in the R2 resection group is presented in Table 3. Representative images of AF are shown in Fig. 1.

Follow-up, recurrence and overall NDS

Five patients were lost to follow-up, and the average follow-up time was 125.6 months (range, 11–524 months). Of the 41 patients with follow-up data, 26 had at least 1 recurrence, and the overall recurrence rate was 63.4% (26 out of 41). For the R0, R1, and R2 resection groups, the recurrence rates were 6.7%, 92.9%, and 100%, respectively. Paired comparisons showed that the R0 resection group had a significantly lower recurrence rate than the other 2 groups (both P values < 0.001). The recurrence rates of the R1 and R2 groups did not differ significantly ($P = 0.538$). In the 14 patients having R0 resection for the first time at our hospital, there were no recurrences. Kaplan–Meier survival analysis indicated significant differences in NDS time among the 3 groups (log-rank test, $P < 0.001$; Fig. 2). The NDS time of the R0, R1, and R2 groups was 80.3 ± 64.8 , 23.6 ± 38.7 , and 9.8 ± 10.8 months, respectively. A one-way ANOVA revealed a significant difference in the NDS time among the 3 groups ($P < 0.001$). Post-hoc comparisons showed that the NDS time of the R0 group was significantly longer than that of the other 2 groups (R0 vs R1, $P = 0.005$; R0 vs R2, $P = 0.001$). However, there was no significant difference in NDS time between the R1 and R2 groups ($P = 0.722$).

Radiotherapy, resection type, and recurrence

Patients who received surgery alone had a significantly lower recurrence rate than those who received surgery

Table 3

Distribution of tumor location in R2 resection group.

Axillary plexus and vessels at the thoracic outlet	30.3%
Paravertebral and costovertebral joints	27.3%
Cervical-thoracic junction	18.2%
Ribs	12.1%
Scapula and surrounding structures	6.1%
Mediastina	6.1%

and radiotherapy (48.1% vs. 88.9%, respectively, $\chi^2 = 7.8$, $P = 0.005$). The interval from surgery to first recurrence in the patients having surgery alone and in those receiving surgery plus radiotherapy was 13.7 ± 10.4 and 11.1 ± 8.5 months, respectively ($t = 0.751$, $P = 0.459$). We also found that there were no significant differences in the recurrence rates in patients having surgery alone in comparison with those receiving surgery and radiotherapy within each resection type (R0, 0% vs 33.3%, $P = 0.188$; R1, 90% vs 100%, $P = 0.667$; R2, 100% vs 100%, $P = 1.0$).

Chemotherapy and recurrence

Four of the five patients receiving postoperative chemotherapy relapsed. Thus, the recurrence rate was 80%. These four patients had margin-positive surgery (i.e., one had an R1 resection and three had R2 resections). Of the three patients having R2 resections, one received prophylactic chemotherapy. One of the five patients who received an R0 resection and postoperative chemotherapy remains asymptomatic without any signs of recurrence.

Discussion

In this large series of patients with chest AF, we found that R0 resection was associated with a lower recurrence rate and a longer NDS time. In addition, our study results showed that additional postoperative radiotherapy did not reduce recurrence rates. Furthermore, the time from surgery to relapse of the patients receiving surgery plus radiotherapy was not longer than that of the patients having surgery alone. We also observed that chemotherapy may not be useful for preventing relapse.

In the present analysis of 41 patients with chest AF, the recurrence rate was 6.7% for R0 resection, 92.9% for R1 resection, and 100% for R2 resection. In an earlier study conducted by Lev et al., the recurrence rate for margin-negative resection (16%) was also lower than that for margin-positive resection (19%).⁹ In a retrospective study including 151 Chinese patients with desmoid tumors, Huang and colleagues indicated that margin status was a strong prognostic factor for recurrence-free survival, and the risk of margin-positive resection was three times higher than that of margin-negative resection.¹⁰ A similar outcome was also reported by Huang and Tzen, who found that a

Table 2

Excision areas and methods of chest wall reconstruction.

Excision areas	Methods of chest wall reconstruction			
	Direct suture	Surgical patch	Metallic internal fixation	Myocutaneous flap
Chest wall soft tissue	51	Nil	Nil	1
Bony chest wall	10	4	2	Nil
Skin scar and extensive full-thickness chest wall resection	1	1	2	7
Extensive resection including diaphragmatic pericardium, vessels, nerves, joints, and other important structures	4	1	1	Nil

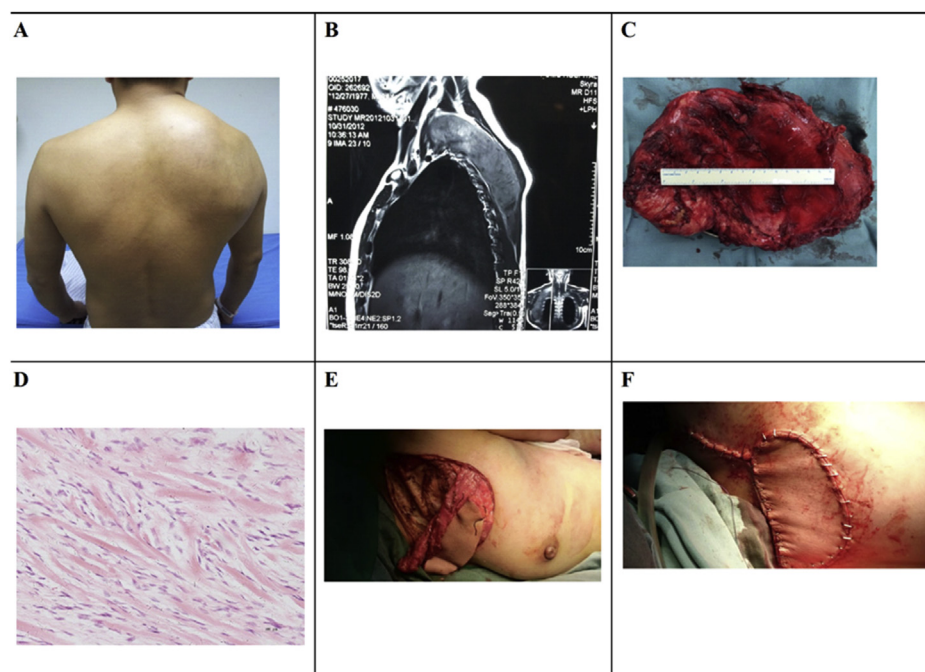


Figure 1. (A–D) A 34-year-old man with chest aggressive fibromatosis (AF) in the scapular region. (A) Preoperative image. (B) Magnetic resonance image. (C) Gross specimen measuring $22 \times 15 \times 7$ cm and weighing 1308 g. (D) Abundant collagen is present between cells (H&E, $\times 20$). (E, F) Extended resection of an AF in the left subclavian region and reconstruction using a latissimus dorsi myocutaneous flap to repair the chest defect in a 31-year-old woman.

positive surgical margin was the only significant factor associated with recurrence of AF.¹¹

Examination of tumor locations for the cases having R2 resection in this study indicate that resection involving the axillary plexus and vessels, paravertebral and costovertebral joints, and cervical-thoracic junction is often restricted. Based on the biological hypothesis, a positive margin resection can result in cells with possible gene mutations being placed in a tissue wound healing environment, which might act as a trigger for relapse.⁵ However, contradictory

findings have been reported in the literature. Salas et al. found that surgical margins were not independently associated with disease progression in multivariate analysis.¹² As indicated by Al-Jazrawe et al., the differences in patient clinical characteristics, in the severity of tumor progression, and in the definition of negative margins might explain the lack of consistencies between research findings.¹³ As several recent studies have pointed out,^{8,13} we consider that a wide R0 resection preserving as much function as possible is still a valid choice for symptomatic chest AF under the circumstances where the observational approach is not suitable.

Due to slow regression, preoperative (neoadjuvant) radiotherapy of AF is not possible to guarantee a microscopically negative resection.¹⁴ With regard to postoperative radiotherapy, the indication is primarily for high-risk patients (i.e., those with a positive, unclear, or narrow margin).² In a retrospective study of 119 AF patients conducted by Shin and colleagues, 29 patients developed postoperative recurrences.¹⁵ Among the 29 patients, the researchers found that the time interval between surgical resection and tumor recurrence of the 11 patients receiving adjuvant radiotherapy was significantly longer than that of the other 18 patients having surgical resection only. However, the current study results showed that there was no association between adjuvant radiotherapy and the relapse interval. That is, we found that postoperative radiotherapy did not delay the recurrence of AF after surgical resection. Even within each resection type, adjuvant radiotherapy did not reduce the recurrence rate.

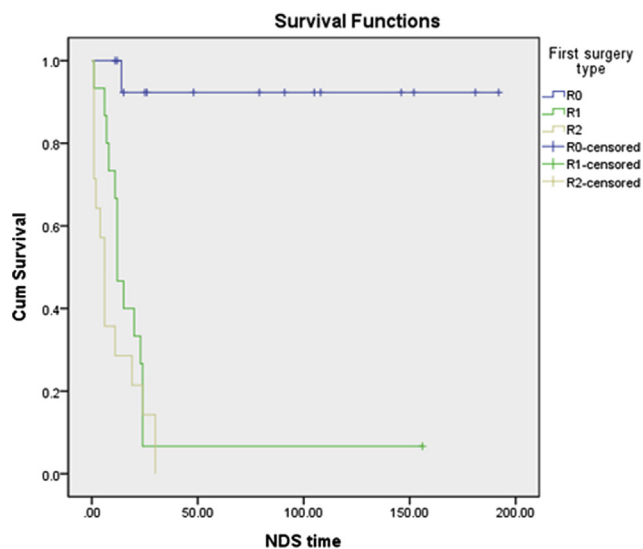


Figure 2. Kaplan–Meier curves of non-disease survival time ($n = 41$).

As indicated by Huang et al., the efficacy of radiotherapy as postoperative treatment for AF has been controversial, and both better and worse results with adjuvant radiotherapy were quoted.¹⁰ Because there is a lack of large, prospective randomized trials comparing the efficacy between surgical resection with or without radiotherapy, it is not possible to derive a firm conclusion regarding the role of radiotherapy in the management of AF. Furthermore, due to radiation-associated morbidity, the pros and cons need to be weighed before the application of radiotherapy.¹³

Chemotherapy is one of the systemic treatments of AF that has been used as an adjuvant treatment in patients who are not suitable for complete resection, or in patients with life- or function-threatening clinical conditions due to the location or size of the tumor.^{13,16} A few recent studies have confirmed that chemotherapy is an efficient therapeutic option. In a retrospective study conducted by the French Sarcoma Group, the researchers found that the median progression-free survival achieved by chemotherapy was 40.8 months in 62 patients with progressive or recurrent desmoid tumors.¹⁷ However, this study also pointed out that after the completion of chemotherapy, there were 29 patients receiving additional intervention for progressive and/or recurrent disease, and another 29 patients developed disease progression after a mean time of 36 months. That is, there were 58 (93.5%) patients developing recurrences. In the current study, we found that the recurrence rate was 80% in patients receiving chemotherapy. Because the only patient who remains asymptomatic received R0 resection and the remaining patients had R1 or R2 resections, we consider that incomplete resection might be more associated with disease recurrence. In our opinion, despite the fact that chemotherapy can delay the time to recurrence, the recurrence rate is not reduced. Due to a small sample size of this subgroup of patients, our hypothesis needs to be further examined in future research.

We suggest that in patients with recurrences, after evaluation of surgical resectability by MRI, reoperation should be conducted with the goal of achieving negative margins. If AF in the subclavian area invades the cervicothoracic junction, the sternoclavicular joint should be opened to resect the tumor if necessary. If a bone scan shows increased radionuclide uptake in the bone surrounding the AF and imaging examination suggests bone invasion, en-bloc resection is required for the involved bony chest wall. Bony reconstruction is required if more than 2 ribs are removed or for cases with sternal and clavicular defects. The commonly used materials include polypropylene mesh, Gore-Tex patch, bone cement, titanium plate or mesh, and a spinal internal fixation system. Soft tissue reconstruction is often performed by transferring a pectoralis major myocutaneous flap, a latissimus dorsi myocutaneous flap, or a rectus abdominis myocutaneous flap of the patient's own tissue.

There are limitations of this study that must be considered. The study was retrospective; thus, there is inherent difficulty in controlling for patient selection for surgery.

In addition, the prognosis of AF in relation to molecular factors cannot be addressed in the current study due to a lack of data. The small sample size of the subgroup of patients receiving chemotherapy also precludes a thorough evaluation of its effectiveness.

Conclusions

Taken together, the current study results suggest that R0 resection is the most effective treatment for chest AF and provides markedly improved outcomes, including a significantly lower recurrence rate and a statistically longer NDS time, compared to R1 and R2 resections. We also found that radiotherapy does not postpone the time from surgical resection to recurrence. Neither radiotherapy nor chemotherapy had an effect on reducing recurrence rates.

Conflict of interest statement

All authors report no potential conflicts of interest.

Acknowledgment

None declared.

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